

SUBACUTE TOXICITY STUDIES OF AN ANTIMICROBIAL COMPOUND ISOLATED FROM A *STREPTOMYCES* SPECIES

ZAKIA SULTANA SATHI, MD. AZIZ ABDUR RAHMAN AND M. A. GAFUR*

Department of Pharmacy, University of Rajshahi, Rajshahi-6205, Bangladesh.

Phone: 880-721-750041, Ext.: 4110, Fax: 880-721-750064

Email: rajucc@citechco.net

ABSTRACT

The subacute toxicity study of an antimicrobial metabolite, 4-hydroxy nitrobenzene (ZS-3), isolated from a *Streptomyces* species, was carried out on long Evan's rats. The studies includes: the gross general observation such as changes in body weight, haematological profiles such as total count of RBC and WBC, differential count of WBC, platelet count and hemoglobin percentage; biochemical parameters of blood such as SGOT, SGPT, serum alkaline phosphatase, uric acid and creatinine and histopathology of the liver, kidney, heart, lung and spleen of both control and experimental groups of rats. The change of haematological and biochemical parameters were statistically insignificant. No abnormalities were found in the histopathology of the liver, kidney, heart, lung and spleen in the experimental group of rats at a dose of 300µg/rat/day for 14 consecutive days when compared with the control group.

INTRODUCTION

Discovery of penicillin by Sir Alexander Fleming in 1929 and its therapeutic application by Professor Flory and Chain in 1941 opened the research area of antibiotics used in the treatment of different micro- bial diseases. With regards to the number of antibiotics produced, the bacterial order actinomycetales, in particular the genus of *streptomyces* occupies the first position (Waksman and Woodruff, 1990). In recent years, owing to indiscriminate use of antibiotics and other unknown reasons, the pathogenic organisms are gaining resistance to existing antimicrobial agents. Hence, the search for the isolation of new, more potent and safer antimicrobial compounds are going on and at the same time many of the marketed antibiotics are being withdrawn due to the serious side effects (Roche, 1950).

With this concept, a strain of Actinomycetes was isolated from a soil sample of Rajshahi, Bangladesh and was identified as *streptomyces* species (Holt *et al.*, 1994). From the broth culture of this organism, an antimicrobial compound, isolated by solvent extraction and preparative thin layer chromatographic technique (Touchston and Dobbins, 1978) was identified as 4- hydroxy nitrobenzene by spectral analysis (Sathi, 2000).

Since almost all of the drugs show unavoidable toxic effects, therefore, in order to assess the safety and efficacy of a drug, toxicity studies should be carried out in animals like mice, rats, guineapigs, dogs and monkeys etc. We herein report the toxicological study of the compound, 4-hydroxynitrobenzene at a dose of 300µg/rat/day on long evan's rats for 14 consecutive days.

*Correspondence

MATERIALS AND METHOD

Collection, maintenance and grouping of experimental rats:

For the purpose of study, 8 long Evan's rats of same sex (male) and age (adult) were collected from ICDDR, Mohakhali, Dhaka.

The rats were kept in properly numbered iron cages individually in a clean animal house with an optimal room temperature (25-30°C) and were given ideal food (Hawk *et al.*, 1954). The animals were maintained in

µl) and antibiotic (ZS-3), 300 µg/rat/day respectively (Table-1).

Gross general observation after drug administration:

During the whole experimental period their behavior, CNS excitation, CNS depression, reflexes, muscular weakness, salivation, diarrhea and food intake were observed. The body weight of each rat of group A and B were measured before administration of the drug and after the completion of the treatment prior to scarify

Table 1
Effect of antibiotic (ZS-3) on body weight of rats

Group	Dose (i.p.) 300µg/ rat/day	Body weight (gm) before drug treatment N= 4, M ₁ ± SD ₁	Body weight(gm) after drug treatment N= 4, M ₂ ± SD ₂	Perce- tage	Calculated t value	t value at 5% level of significance	Re-marks
A	300 µl vehicle	106.5 ± 2.65 (103,106, 108,109)	115 ± 7.12 (109,115,111, 125)	+ 7.98	+ 2.24	2.447	NS
B	300 µg comp.	115.5 ± 6.56 (110,125, 113,114)	121 ± 5.10 (116,128,121, 119)	+ 4.76	+ 1.32	2.447	NS

NB: M₁ and M₂ = Sample mean value, SD₁ and SD₂ = Standard deviations, n = Number of rats,
+ = Increase, NS = Not Significant, i.p. = intraperitoneally.

this way for 15 days before drug administration and continued up to the end of the experiment. Individual weight of the rats was taken and were grouped into two groups. The rats of group B (average weight 115.5g) were used for experiment while that of group A (average weight 106.5g) were used as control.

Sample administration:

The antibiotic (ZS-3) was dissolved in distilled water using tween-20 as cosolvent, so that 0.3 ml contained 300 µg of the antibiotic. The rats in group A and B were injected intraperitoneally with vehicle (300

them.

Procedure for checking the haematological profiles of blood:

For haematological studies, blood was drawn from the tail veins of all the rats in group A and B before the commencement of drug administration. Then blood smears were made on glass slides and stained with "Leishmen reagent" to perform TC (Total count), DC (differential count) and platelet count. With the use of capillary tubes blood was drawn from each of the rat to estimate the hemoglobin percentage by "Van Kampen-Zijlstra's" method. This was

the pre-haematological study on normal rats. Post haematological studies were done on 7th and 14th days after the commencement of

drug administration following the same procedure used for normal rats.

Table - 2
Hematological Profiles of Group A rats (control)

Hematological parameters		Normal rats	Rats treated with vehicle	
		1 st day	7 th day	14 th day
(i) Total RBC count (million/cu.mm)		4.8	5.8	4.9
		4.6	5.6	5.0
		4.9	5.8	5.2
		5.1	5.3	4.8
		4.85±0.210	5.63±0.236	4.97±0.17
(ii) Total WBC count (no/cu. mm)		11600	10800	11200
		11400	10500	10700
		11300	11000	11100
		11300	10900	10800
		11400±141	10800±216	10950±238
(iii) Differential count of WBC	a. Neutrophil	38	36	37
		39	35	39
		41	37	38
		40	38	37
		39.5±1.29	36.5±1.29	37.75±0.957
	b. Lymphocyte	53	54	52
		52	54	55
		53	55	54
		51	56	54
		52.25±0.957	54.75±0.957	53.75±1.26
	c. Monocyte	5	6	6
		5	6	3
		3	4	4
		5	3	5
		4.5±1.0	4.75±1.50	4.5±1.29
	d. Eosinophil	4	4	5
		4	5	3
		3	4	4
		4	3	4
		3.75±0.50	4±0.816	4±0.816
(iv) Platelet count (no/cu. mm)		335000	300000	340000
		325000	350000	290000
		320000	340000	310000
		310000	340000	300000
		322500±10408.33	332500±22173.56	310000±21602
(v) Hemoglobin (%)		13.2	13.6	12.8
		13.8	13.9	13.7
		13.8	13.8	13.3

Table - 3
Hematological Profiles of Group B rats

Hematological parameters		Normal rats	Rats treated with ZS-3	
		1 st day	7 th day	14 th day
(i) Total RBC count (million/cu.mm)		4.9 5.0 5.2 4.8 4.97±0.17	4.6 4.5 4.8 5.0 4.73±0.22	5.3 5.8 5.6 5.8 5.63±0.23
(ii) Total WBC count (no/cu. mm)		12200 12400 12800 12500 12475±250	13000 12500 12500 12800 12700±244.95	12400 12800 12500 12600 12575±170.78
(iii) Differential count of WBC	a. Neutrophil	40 45 43 42 42.5±2.08	43 41 42 40 41.5±1.29	40 41 45 40 41.5±2.38
	b. Lymphocyte	49 45 47 49 47.5±1.94	50 49 47 51 49.25±1.70	50 51 48 51 50±1.41
	c. Monocyte	6 6 6 5 5.75±0.5	4 5 6 5 5±0.816	6 4 4 5 4.75±0.96
	d. Eosinophil	5 4 4 4 4.25 ± 0.5	3 5 5 4 4.25 ± 0.96	4 4 3 4 3.75± 0.5
(iv) Platelet count (no/cu. mm)		340000 320000 350000 350000 340000 ± 14142	290000 300000 300000 300000 297500 ± 5000	320000 320000 310000 300000 312500 ± 9574
(v) Haemoglobin (%)		13.0 13.3 12.7 12.2 12.8± 0.47	12.8 12.6 13.2 13.3 12.98± 0.33	13.4 13.4 13.4 12.8 13.25± 0.3

Procedure for checking the biochemical parameters of blood:

For the study of biochemical parameters the rats of all groups were sacrificed with the help of a surgical blade No. 22 on the 14th day of treatment with ZS-3, and the blood were collected in plastic centrifuge tubes. These were then allowed to clot at 40°C for 4 hour. After clotting, the blood samples were centrifuged at 4000 rpm for 15 minutes using

a WIFUNG centrifuge LABO-50M. The clear straw color serum was then collected in vials with Pasteur pipette and stored at 20°C. Then the enzymes SGOT (Serum glutamate oxaloacetate transaminase), SGPT (Serum glutamate pyruvate transaminase), serum alkaline phosphatase and serum creatinine, uric acid and urea were determined by using the procedures and reagents described in "Enlehringer Mannheim GmbH Diagnostica"

(king and Armstrong, 1934; Reiteman and Frankel, 1957; Fawcett and Scott, 1960; Coulombe and Favreau, 1963).

Procedure for the histopathological study of liver, kidney, heart, lungs and spleen:

The liver, kidney, heart, lungs and spleen of all of the rats of group A and B were collected after sacrificing them at the 14 th

mounted on glass slides with diphenyl xylene mounting fluid and observed under microscope at the “Bangladesh Sericulture Research Institute”, Rajshahi, Bangladesh.

RESULTS AND DISCUSSIONS

Gross general observation:

The rats of group A and B were being

Table - 4
Effect of antibiotic (ZS-3) on rats blood after i.p. administration of 300 µg/rat/day for 14 consecutive days

Biochemical Parameters	Group A Control rats n=4, M ₁ ± SD ₁	Group B n=4 M ₂ ± SD ₂	% change	Calculate d t value	t value at 5% level of significance	Re- marks
SGOT (IU/L)	11.75 ± 1.25 (13,12,12,10)	11.5 ± 1.29 (12,11,10,13)	-2.13	0.278	2.447	NS
SGPT (IU/L)	7.25 ± 1.5 (9,6,6,8)	8.25 ± 1.26 (10,8,8,7)	+13.79	1.021	2.447	NS
SALP (IU/L)	44.25 ± 3.30 (48,45,44,40)	41.25 ± 1.89 (44,41,40,40)	-6.78	1.578	2.447	NS
Serum uric acid (mg/dl)	7.3 ± 0.22 (7.5,7.4,7.3,7.0)	7.05 ± 0.129 (7.1,6.9,7.0,7.2)	-3.42	1.961	2.447	NS
Urea (mg/dl)	41.5 ± 2.64 (45,42,40,39)	42.25 ± 1.708 (44,43,42,40)	+1.8	0.477	2.447	NS
Serum creatinine (mg/dl)	1.07 ± 0.188 (1.25,1.2,0.98, 0.85)	0.85 ± 0.05 (0.9,0.85,0.8, 0.75)	-20.63	2.272	2.447	NS

Table – 5
Histopathological studies after treatment with antibiotic ZS-3 at a dose level of 300µg/rat/day for 14 consecutive days.

Group	Dose (i.p) µg/rat/day	Histopathological changes observed				
		Heart	Lungs	Spleen	Liver	Kidney
A	300µl vehicle	NAD	NAD	NAD	NAD	NAD
B	300µg	NAD	NAD	NAD	NAD	NAD

NAD = No abnormality detected.

day of observation. The tissues were sliced into pieces and immersed in 10% formaline for three days, processed, stained with “Harris Hematoxilin and eosin reagent”,

treated with vehicle and ZS-3 respectively, showed no signs of tremor, convulsions and reflex abnormalities. The body weights of all the rats (both group A and B) were increased

after treatment. Moreover, no muscular numbness of the hind and fore legs, salivation or diarrhea was observed. The food intake per day was also found normal. So, from the results, it is decided that drug has no effect on normal growth. The results are shown in Table-1.

Haematological profiles:

Haematological profiles were studied on normal rats (before treatment) and after 7 and 14 days of treatment. Each time the value of the parameters in each rat was changed slightly. However, the parameters remained within the normal range. The results of hematological profiles are shown in Table-2 and Table-3.

Biochemical Parameters:

Biochemical parameters were studied on normal rats (before treatment) and after 7 and 14 days of treatment. Each time the value of the parameters in each rat was changed slightly. However the parameters remain within the normal range. This indicates that the antibiotic has no adverse effects on liver and kidney function (Table-4).

Histopathological Studies:

At the 14th day of drug treatment and observation, the animals of both control and experimental groups were sacrificed and the organs such as liver, kidney, lung, spleen and heart were isolated and histopathological examinations were done. No detectable abnormality in the histopathological point of view was observed between the control and the drug treated rates when the tissue slides were examined under microscope. This indicates that the compound has no effects on cellular structure, i.e. the compound does not cause degeneration of cells of these organs (Table-5).

ACKNOWLEDGEMENTS

The authors like to thank Dr. Naoki Sugimoto, National institute of Health Sciences, Tokyo, Japan for spectral analysis

and Dr. Anwarul Habib, Rajshahi Medical Collage, Rajshahi, Bangladesh for toxicological studies.

REFERENCES

- Coulombe, J.J. and Favreau, L. (1963). A new simple semimicro method for determination of Urea. *J. clin. chem.* **9**: 102-108.
- Fawcett, J.K. and Scott, J.E. (1960). A new simple semimicro method for determination of Urea., *J. Clin. Path.* **13**: 156-159.
- Hawk P.B., Oser L., and Summerson W.H. (1954). Practical Physiological chemistry 13th edition McGraw Hill Book Company U.S.A.
- Holt, J.G., Krieg, N.R., Sneath, P.H.A., Staley, J.T., Williams, S.T. (1994). Bergey's Manual of determinative Bacteriology 9th Edition, P 307-308.
- King P.J.E. and Armstrong A.R. (1934). A convenient method for determination of serum and bile phosphate activity. *Cand. Med. Assoc.* **31**: 376-381.
- Reitman, S. and Frankel, S. (1957). A colorimetric method for the determination of serum glutamate-oxalate and glutamate-pyruvate transaminases. *Am. J. Clin. Path.* **28**: 56-63.
- Roche, W. (1950). New antibiotics. *Pharmazee*. **5**:190.
- Sathi, Z. S. (2000). Chemical and Biological studies on Metabolites of *streptomyces* and *Nocardiopsis species* M. Pharm. thesis submitted to the Dept. of Pharmacy, University of Rajshahi, Bangladesh.
- Touchston, J.C. and Dobbins, F.M. (1978). Practice of thin layer chromatography, 1st edition, John Willey and sons co. Ltd.
- Waksman, S.A. and Woodruff, H.B. (1990). Bacteriostatic and bactericidal substances produced by a soil action-mycetes, Biomedical health publication. **45**: 609-614.